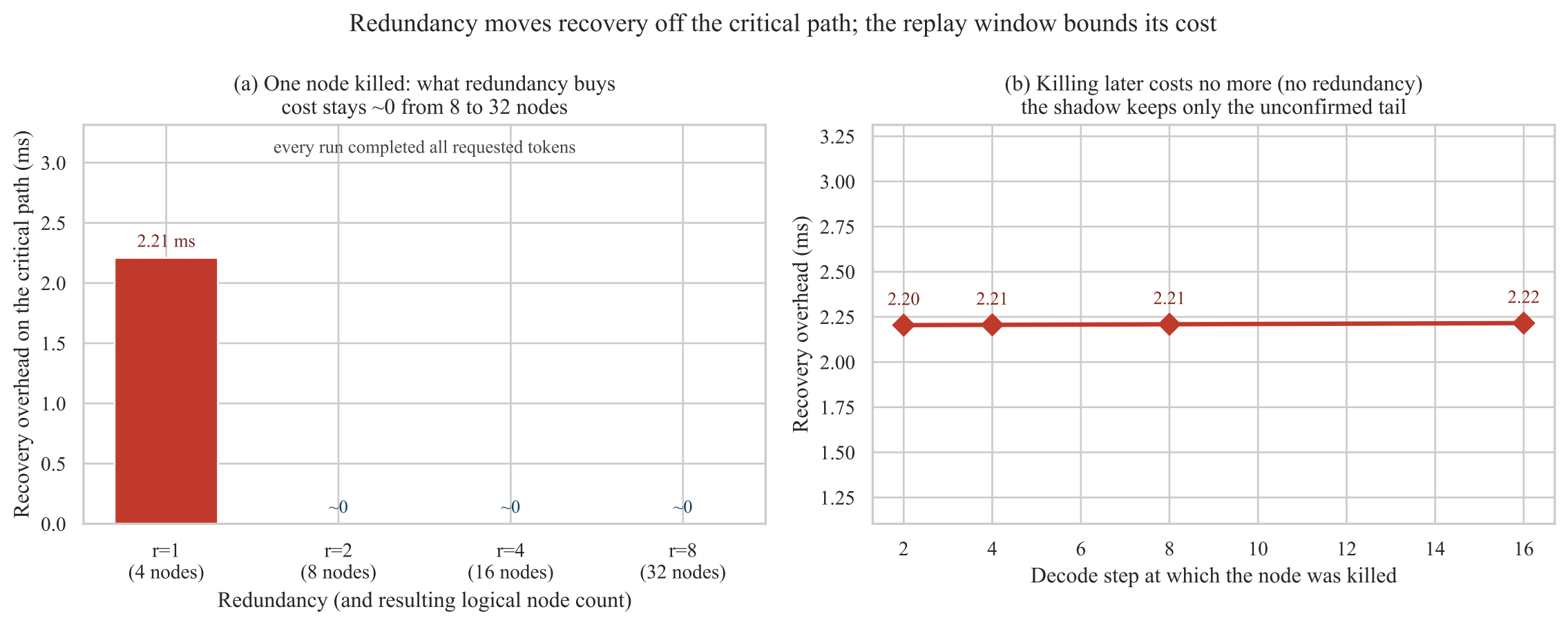
Redundancy moves recovery off the critical path; the replay window bounds its cost
(a) One node killed: what redundancy buys
cost stays ~0 from 8 to 32 nodes
every run completed all requested tokens
Recovery overhead on the critical path (ms)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
2.21 ms
~0
~0
~0
r=1
(4 nodes)
r=2
(8 nodes)
r=4
(16 nodes)
r=8
(32 nodes)
Redundancy (and resulting logical node count)
(b) Killing later costs no more (no redundancy)
the shadow keeps only the unconfirmed tail
Recovery overhead (ms)
3.25
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
2.20
2.21
2.21
2.22
2
4
6
8
10
12
14
16
Decode step at which the node was killed